Compton scattering

Nazim K and Daniel S

August 2026

Compton scattering is the scattering of photons off electrons. Let the photon be incident along the horizontal with wavelength λ and scatter off at an angle θ above the horizontal with a new wavelength λ' with the previously stationary electron scattered at speed v at angle ϕ below the horizontal. Our aim is to compute λ', v and ϕ given λ and θ. To conserve energy and momentum we must first relate them to wavelength. We need Planck's formula:

$$E = hf = \frac{hc}{\lambda}$$

We already saw how this formula arises from treating light as a quantum field in the section on Black bodies. Here the revelation will be that photons have momentum. Since the photon moves at c, unfortunately, we cannot use Newton's laws like for billiard ball scattering but require special relativity. One helpful formula is the energy momentum invariant: $E^2 - p^2c^4 = m^2c^2$ which is, well, invariant. Another is the formula for total energy: $E = \gamma mc^2$ and total momentum: $p = \gamma mv$, ($\gamma = \frac{1}{1-\frac{v^2}{c^2}}$ is the lorentz factor.) From here on out we will set c = 1 to make the equations look nicer and will reinsert it at the end by matching dimensions.

We can actually derive the planck formula using the energy momentum invariant and de-Broglie's formula for a photon's momentum $p = \frac{h}{\lambda}$. Photon mass is zero so:

$$E = pc = \frac{hc}{\lambda} = hf$$

This seems fitting since de-Broglie's formula explicitly links a particle like concept: momentum, to a wave-like one: wavelength and Compton scattering highlights this since light scatters like a particle but also experiences a wavelength shift.

Call the photon's momentum before the collision p and after p', with the electron momentum before being zero and p_e after. The cosine rule gives:

$$p_e^2 = p^2 + p'^2 - 2pp' \cos\theta$$

Now for energy and momentum. The electron's energy momentum invariant is:

$$\begin{aligned} m^2 &= E^2 - p_e^2 \\ &= \frac{m^2}{1 - v^2} - p_e^2 \end{aligned}$$

where we have used the formula for total energy and $m = m_e$, the electron's mass. Conserving energy gives:

$$m_e + \frac{h}{\lambda} = \frac{h}{\lambda'} + E$$

When c = 1 the photon's energy and momentum are conveniently the same.

$$m_e + p = p' + E$$

Now we have some equations. It is very easy to go round in circles subbing in different things and eliminating others. First let's try get the shift in wavelength $\Delta\lambda = \lambda' - \lambda$ and so eliminate v and then p_e. Subbing energy conservation into the electron's energy momentum invariant:

$$m^2 = (m + p - p')^2 - p_e^2$$

$$p_e^2 = (p - p')^2 + 2m(p - p')$$

Equating this to the original expression we had for p_e^2 gets rid of p_e.

$$\begin{aligned} p^2 + p'^2 - 2pp' \cos\theta &= (p - p')^2 + 2m(p - p') \\ 2m(\frac{h}{\lambda} - \frac{h}{\lambda'}) + 2(1 - \cos\theta)\frac{h^2}{\lambda\lambda'} &= 0 \\ \frac{h(1 - \cos\theta)}{m} &= \lambda\lambda'(\frac{1}{\lambda} - \frac{1}{\lambda'}) = \Delta\lambda \end{aligned}$$

This is a crucial result so we will return the speed of light to make it SI. Planck's constant has dimensions of energy time: ML^2T^{-1} and so if we divide it by mc the result will have units $M^{1-1}L^{2-1}T^{-1+1} = L$ as required for a wavelength.

$$\Delta\lambda = \frac{h(1-\cos\theta)}{mc}$$

This means we now know

$$\lambda' = \lambda + \frac{h(1-\cos\theta)}{mc}$$

Now to find ϕ. The sine rule gives:

$$p_e \sin\phi = p' \sin\theta$$

Momentum conservation horizontally gives:

$$p = p' \cos\theta + p_e \cos\phi$$

combining these:

$$\tan\phi = \frac{p' \sin\theta}{p - p' cos\theta}$$

note that $\frac{p}{p'} = \frac{\lambda'}{\lambda} = 1 + \frac{h(1-\cos\theta)}{mc}$. Thus:

$$\tan\phi = \frac{\sin\theta}{(1-\cos\theta)(1+\frac{h}{mc})}$$

Now to find v:

$$E = \frac{mc^2}{\sqrt{1-\frac{v^2}{c^2}}}$$

but from conservation of energy, and this being the energy of the electron after the collission,

$$mc^2 + \frac{hc}{\lambda} = \frac{mc^2}{\sqrt{1-\frac{v^2}{c^2}}} + \frac{hc}{\lambda'}$$

so after some rearranging:

$$v = c\sqrt{1 - \frac{m^2c^2}{h^2}(\frac{1}{\lambda} - \frac{1}{\lambda'})^{-2}}$$